

Oracle Database Administration

Technical Standard Operating Procedure: Control File Multiplexing

Database Name (DB_NAME): PROD

Instance SID (SID_NAME): PROD

Objective: High Availability / Redundancy

Environment: Linux / Oracle Database

Overview: Multiplexing control files creates identical concurrent copies on separate physical disk locations, protecting the database instance from disk corruption and complete failure.

Step 1: Check Current Control File Locations

Connect via SQL*Plus and verify existing active control file paths.

```
SQL> SELECT name FROM v$controlfile;
```

```
NAME -----  
/u01/app/oracle/oradata/PROD/control01.ctl /u01/app/oracle/fast_recovery_area/PROD/control02.ctl
```

Step 2: Shutdown the Database

Perform an immediate shutdown to cleanly close and dismount database structures prior to file copying.

```
SQL> SHUTDOWN IMMEDIATE;
```

```
Database closed. Database dismounted. ORACLE instance shut down.
```

Step 3: Copy an Existing Control File to the New Location

Use OS commands to duplicate an existing control file to your secondary volume or target path.

```
[oracle@localhost PROD]$ pwd /u01/app/oracle/oradata/PROD [oracle@localhost PROD]$ cp  
/u01/app/oracle/oradata/PROD/control01.ctl /u01/app/control03.ctl
```

Step 4: Startup Nomount and Update the CONTROL_FILES Parameter

If using **SPFILE**: Start the instance in NOMOUNT mode before altering system parameter settings.

```
SQL> STARTUP NOMOUNT; SQL> ALTER SYSTEM SET CONTROL_FILES =  
'/u01/app/oracle/oradata/PROD/control01.ctl',  
'/u01/app/oracle/fast_recovery_area/PROD/control02.ctl', '/u01/app/control03.ctl' SCOPE=SPFILE;
```

```
ORACLE instance started. System altered.
```

If using a **PFILE** (`init.ora`): Open your text initialization file and update the parameter entry manually:

```
control_files = ( '/u01/app/oracle/oradata/PROD/control01.ctl',  
'/u01/app/oracle/fast_recovery_area/PROD/control02.ctl', '/u01/app/control03.ctl' )
```

Step 5: Restart the Database and Verify

Restart the instance normally to mount and open all multiplexed control files simultaneously.

```
SQL> SHUTDOWN IMMEDIATE; SQL> STARTUP;
```

```
ORACLE instance started. Database mounted. Database opened.
```

Verification Query

Confirm that all three target control files are active and synchronized.

```
SQL> SELECT name FROM v$controlfile;
```

```
NAME -----  
/u01/app/oracle/oradata/PROD/control01.ctl /u01/app/oracle/fast_recovery_area/PROD/control02.ctl  
/u01/app/control03.ctl
```

Execution Complete:

The control file multiplexing procedure including STARTUP NOMOUNT has been successfully applied and verified.